RE-LIV *oral lotion*

Description: White flavoured oral lotion

Content: Lignocaine hydrochloride 2.50%w/v, chlorhexidine hydrochloride 0.50%w/v and triamcinolone acetonide 0.10%w/v.

Indication: For the relief of discomfort and pain, reduction of inflammation and to promote healing in the following conditions: ulcer, lesions and soreness in the gums, tongue, lips or palate; trauma or lesions in the oral cavity caused by new denture, denture sore spots and denture stomatitis; and teething disorders.

Pharmacology: Triamcinolone acetonide - high potency glucocorticoid acts by diffusing across cell membranes and stimulates transcription of mRNA resulting in protein synthesis of various enzymes ultimately responsible for its anti-inflammatory effects. Skin reactions e.g. irritation and pruritis may occur; contact dermatitis reported. Chlorhexidine - bactericidal to G+ve and G-ve bacteria. It disrupts the plasma membrane of the bacterial cell and cellular contents are lost. Lignocaine - local anesthetic for application to mucous membranes.
The absorption of topically applied triamcinolone acetonide depends on the physiological (e.g. degree of hydration) and anatomical (e.g. thickness) characteristic of the region of the skin to which it is applied. Once the drug reaches the dermis, it is absorbed into the systemic circulation, particularly when the skin is broken. There is some systemic absorption of topical corticosteroids through the oral mucosa; absorption increases with increased potency. Once the drug is absorbed into the systemic circulation, it is bound to the plasma proteins. It is mainly metabolised in the liver but also in the kidney and is excreted in the urine. Lignocaine hydrochloride is readily absorbed through mucous membranes into the systemic circulation. The rate of absorption is influenced by the vascularity or rate of blood flow at the site of application and the total dosage administered. Lignocaine undergoes first-pass metabolism in the liver and bioavailability is about 35% after oral administration. Metabolism in the liver is rapid and approximately 90% of a given dose is dealkylated to form monoethylglycinexylidine (MEGX) and glycinexylidide (GX) Both of these metabolism may contribute to the therapeutic and toxic effects of lignocaine. Further metabolism occurs and metabolites are excreted in the urine with less than 10% of unchanged lignocaine. Lignocaine readily crosses the placenta and blood-brain barrier; it is excreted in breast milk.

Adverse reactions: Signs of infection such as pain, redness or pus-containing blister and/or signs of irritation such as burning, itching, blistering or peelings not present before therapy, may require medical attention. Systemic absorption due to long term use may cause systemic corticosteroid effects including Cushing's syndrome, hyperglycaemia, and glycosuria.
Immune system disorders
Frequency not known: Hypersensitivity including anaphylactic shock.

Contraindications: It is contraindicated in herpetic lesions of known viral origin e.g. herpes labialis, intraoral lesions (primary herpetic gingival stomatitis, herpanginas), and in the presence of infection; and known hypersensitivity to any ingredient of the preparation.

Precautions: Care should be taken in patients with tuberculosis, peptic ulceration, or diabetes mellitus, especially long term usage. Since adrenal suppression and growth retardation due to the systemic absorption of topical corticosteroid have been documented in children, special care must be exercised in using corticosteroid in the pediatric patients, especially when extensive areas are treated. If significant regeneration or repair of oral tissues has not occurred in 7 days, additional investigations into the aetiology of oral lesion are advised. It should be borne in mind that the normal defensive responses of the oral tissues are depressed in patients receiving topical corticosteroid therapy. Risk-benefit must be considered in women of childbearing potential and particularly during early pregnancy or nursing mothers. Caution is advised in paediatric, geriatric, acutely ill, or debilitated patients, who may be more susceptible to systemic toxicity of this product. Do not chew gum or food while numbness persists because of risk of biting tongue or buccal mucosa. Do not eat for one hour following use of medication because it may impair swallowing leading to risk of aspiration. This product consists of chlorhexidine hydrochloride and cannot be instilled in to ear.

Re-liv Oral Lotion contains chlorhexidine. Chlorhexidine is known to induce hypersensitivity, including generalised allergic reactions and anaphylactic shock. The prevalence of chlorhexidine hypersensitivity is unknown, but available literature suggests this is likely to be very rare. Re-liv Oral Lotion should not be administered to anyone with a possible history of an allergic reaction to chlorhexidine. If any signs or symptoms of a suspected hypersensitivity reaction such as itching, skin rash, redness, swelling, breathing difficulties, light headedness, and rapid heart rate develop, immediately stop using the product. Appropriate therapeutic countermeasures must be instituted as clinically indicated.

Use in pregnancy and lactation: Pregnancy is not a contraindication, but the user should be aware of possible toxic effects to the foetus if there is systemic absorption. Risk-benefit must be considered in women of childbearing potential and particularly during early pregnancy or nursing mothers.

Drug Interactions: Beta-adrenergic blocking agents: Concurrent use may slow metabolism of lignocaine because of decreased hepatic blood flow, leading to increase to risk of lignocaine toxicity, especially in large doses, repeated administration, oral use (especially if swallowed) of lignocaine. *Cimetidine:* Cimetidine may inhibit hepatic metabolism of lignocaine, leading to increase risk of lignocaine toxicity, especially with large doses, repeated administration, or oral use (especially if swallowed) of lignocaine. Corticosteroid-induced adrenal suppression has been associated not only with systemic therapy, but has followed topical application of corticosteroid preparations, particularly those containing potent corticosteroids. Adrenal suppression has also been associated with the use of inhalants and the topical application of eye drops, eye ointments, and nasal preparations. Systemically absorbed topical corticosteroid may interact with concurrent drug therapy given to patients. NSAIDs: Concomitant administration of ulcerogenic drugs such as indomethacin during corticosteroid therapy may increase risk of gastro-intestinal ulceration. Aspirin should be used cautiously in conjuction with corticosteroids in patients with hypoprothombinemia. Although concomitant therapy with salicylates and corticosteroids does not appear to increase the incidence or severity of gastro-intestinal ulceration, the possibility of this effect should be considered. *Potassium-depleting drugs*: Potassium depleting diuretics (e.g. thiazides, frusemide, and ethacrynic acid) and other drugs that deplete potassium, such as amphotericin B, may enhance the potassium wasting effect of corticosteroid. Serum potassium should be closely monitored in patients receiving corticosteroids and potassium depleting drugs. *Anticholinesterase agents:* Interaction between corticosteroids and anticholinesterase agents such as ambinonium, neostigmine, or pyridostigmine (and presumably organophosphate anticholinesterase pesticides) can produce severe weakness in patients with myasthenia gravis. If possible anticholinesterase should be withdrawn at least 24 hours prior to initiation of corticosteroid therapy.

Dosage and administration: Instill 2 - 4 drops of lotion to the lesion using the tip of a clean cotton bud or finger. Rub gently to cover the lesion with lotion. Application may be repeated after 1 - 2 hours, preferably after meals, and at bedtime.

Treatment of over dosage: Activated charcoal has been suggested for use in the treatment of oral overdosage with lignocaine. The use of gastric lavage has also been suggested but this could be hazardous if convulsion is imminent. Acidification of the urine has been used to increase renal excretion of lignocaine but any increase is likely to be of any significance. Supportive and symptomatic steps should be taken to maintain the circulation and respiration and to control convulsion. Vasopressor agents have been suggested in the treatment of marked hypotension although their use is accompanied by a risk of central nervous system excitation. Vasopressors should not be given to patients receiving oxytocic drugs.

Storage conditions	:	Keep container tightly closed. Store in a dry place below 30˚c. Protect from light. Keep away from children.
Shelf life	:	3 years
Presentation	:	5ml/ bottle
Manufactured & Distributed by	:	**PRIME PHARMACEUTICAL SDN.BHD.** 1505, Lorong Perusahaan Utama 1, Taman Perindustrian Bukit Tengah, 14000 Bukit Mertajam, Penang, Malaysia.

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